VERA-ECG: Verifiable Evidential Reasoning and Anatomy-Grounded Multimodal Framework for Reliable 12-Lead ECG Interpretation

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Yihao Wen: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization.

Jun Zeng: Supervision, Project administration, Resources, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical approval and data availability

All experiments used publicly available, de-identified datasets (PTB-XL, CPSC2018, and ECG-QA). No new human or animal experiments were performed. The data and code availability statement is provided in the manuscript.